

MASTER'S COMPREHENSIVE EXAMINATION

Part II

Applied (90 minutes)

April 10, 2010

Instructions: Attempt any **three** out of the following four questions on both pages. Mention which three you would like to have graded. Open book and open notes.

1. The Kenton Company is testing 4 different package designs for a new breakfast cereal; 20 stores, with about equal sales volumes, are the experimental units. Each store is randomly assigned one package designs, with each package design being assigned to 5 stores. A fire later destroys one store so this store is dropped from the study. Sales, in number of cases, for the study period are recorded in Table 1. The characteristics of 4 package designs are shown in Table 2. The data is analyzed using one-way layout with fixed effects.

Table 1: Number of Cases Sold by Stores for Each of Four Package Designs

	Package Design			
	1	2	3	4
Cases Sold Per Store	11	12	23	27
	17	10	20	33
	16	15	18	22
	14	19	17	26
	15	11		28
Mean	$\bar{y}_1=14.6$	$\bar{y}_2=13.4$	$\bar{y}_3=19.5$	$\bar{y}_4=27.2$
Standard Deviation	2.3	3.6	2.6	4.0

Table 2: Characteristics of Four Package Designs

Package Design	Characteristics
1	3 color, with cartoons
2	3 color, without cartoons
3	5 color, with cartoons
4	5 color, with cartoons

A. Copy the ANOVA table below to your Blue Book and fill in the blanks

One-way ANOVA Table				
Source	Degrees of Freedom	Sum of Squares	Mean Squares	F
Between design	3	_____	_____	_____
Error	_____	_____	_____	
Total	18	_____		

B. Conduct an F test to test at $\alpha = 0.05$ of the null hypothesis that the four package designs have the same sales.

C. Use Tukey's method to perform multiple comparisons of the four package designs at $\alpha = 0.05$

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2. A scientist observes grackles' attempts at mating. In 75 such attempts, 21 are successful. It is well known that successful matings can only result in 1 or 2 offspring. Among the successful attempts, 10 have single offspring born and 11 have two offspring born.
- A. What is the point estimate and 90% large sample confidence interval for the expected number of offspring per mating?
- B. What is a point estimate and 90% large sample confidence interval for the probability that a randomly chosen newborn grackle will have a twin?
3. While trying to recreate a dinosaur, some Rutgers scientists mistakenly created a new animal that is now known as the Piscataway Carnivorous Turtle (PCT). The animals immediately escaped and are now found throughout New Jersey. Weights of PCTs are distributed $N(\mu=50, \sigma=10)$ for males and $N(\mu=45, \sigma=9)$ for females; 60% of PCTs are female.
- a. What is the overall mean and standard deviation of weight for all PCTs?
- b. New Jersey State regulations only allow PCTs that weigh more than 60 to be sampled. If a PCT weighs exactly 60 what is the probability it is a male?
- c. If a PCT weighs > 60 , what is the probability it weighs > 65 ?
4. In a study, 31 of 534 chickens with genotype X are determined to be unfit for human consumption compared to 52 of 1476 chickens that do not have genotype X. Let P_{X+} and P_{X-} be the true probabilities that chickens who do and do not have genotype X, respectively, are unfit to eat.
- A. Do a statistical test of $H_0: P_{X+} = P_{X-}$ at $\alpha = 0.05$
- B. Give a 95% confidence interval for the difference $D_X = P_{X+} - P_{X-}$
- C. Give a 95% confidence interval for the ratio $R_X = \frac{P_{X+}}{P_{X-}}$